

Pandas.ipynb X

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```
!pip install pandas
```

[1]

```
Requirement already satisfied: pandas in c:\users\sai\appdata\local\pandas
Requirement already satisfied: numpy>=1.26.0 in c:\users\sai\appdata\local\pandas
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\sai\appdata\local\pandas
Requirement already satisfied: pytz>=2020.1 in c:\users\sai\appdata\local\pandas
Requirement already satisfied: tzdata>=2022.7 in c:\users\sai\appdata\local\pandas
Requirement already satisfied: six>=1.5 in c:\users\sai\appdata\local\pandas
```

```
[notice] A new release of pip is available: 25.1.1 -> 25.3
```

```
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
import pandas as pd
```

[2]

```
pd.__version__
```

[3]

```
'2.3.3'
```

```
#1. DataFrame
df = pd.DataFrame([1,2,3], columns=['Col_name'])
```

[7]

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#1. DataFrame

```
df = pd.DataFrame([1,2,3], columns=['Col_name'])  
print(df)
```

[7]

	Col_name
0	1
1	2
2	3

```
print(type(df))
```

[8]

<class 'pandas.core.frame.DataFrame'>

```
data = {  
    'Name' : ['Ankit' , 'Pawan' , 'Mayur' , 'Viki'],  
    'Age' : [16 , 17 , 18, 19],  
    'Salary' : [1000000, 90000, 80000, 70000]  
}
```

[9]

```
df = pd.DataFrame(data)
```

[10]

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```
df = pd.DataFrame(data)
print(df)
```

[10]

	Name	Age	Salary
0	Ankit	16	100000
1	Pawan	17	90000
2	Mayur	18	80000
3	Viki	19	70000

2. Basic dataframe Understanding

[11]

```
df.head(2) #top rows
df.tail(2) # bottom
```

[12]

	Name	Age	Salary
2	Mayur	18	80000
3	Viki	19	70000

```
df.shape
```

[13]

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df.shape

[13]

(4, 3)

df.columns

[14]

Index(['Name', 'Age', 'Salary'], dtype='object')

df.rename(columns={'Salary' : 'Package'}, inplace=True)

[18]

df.info()

[19]

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    Name         4 non-null      object
1    Age           4 non-null      int64
2    Package       4 non-null      int64
dtypes: int64(2), object(1)
```

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```
df.describe()
```

[20]

	Age	Package
count	4.000000	4.000000
mean	17.500000	310000.000000
std	1.290994	460072.458061
min	16.000000	70000.000000
25%	16.750000	77500.000000
50%	17.500000	85000.000000
75%	18.250000	317500.000000
max	19.000000	1000000.000000

```
#3. Save and Load Data From CSV
```

[21]

```
df.to_csv('Test_data.csv', index=False) #export data
```

[22]

```
load_df = pd.read_csv('Test_data.csv') #import data
```

[24]

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```
[24] load_df = pd.read_csv('Test_data.csv') #import data
      load_df
```

	Name	Age	Package
0	Ankit	16	1000000
1	Pawan	17	90000
2	Mayur	18	80000
3	Viki	19	70000

4. Rows and Columns - Selection

```
[25] #select columns
      df[['Name']] #single column
      df[['Name', 'Age']] #multi col
```

	Name	Age
0	Ankit	16
1	Pawan	17
2	Mayur	18

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4. Rows and Columns - Selection

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```
#select columns
df[['Name']] #single column
df[['Name', 'Age']] #multi col
```

[25]

	Name	Age
0	Ankit	16
1	Pawan	17
2	Mayur	18
3	Viki	19

```
#select rows
#loc = index name based value leta he
df.loc[df.Name=='Ankit']
df.loc[(df.Name=='Ankit') & (df.Age==16)]
df.loc[0:2]
```

[35]

	Name	Age	Package
0	Ankit	16	1000000
1	Pawan	17	500000

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```
[35] #select rows  
#loc = index name based value leta he  
df.loc[df.Name=='Ankit']  
df.loc[(df.Name=='Ankit') & (df.Age==16)]  
df.loc[0:2]
```

	Name	Age	Package
0	Ankit	16	1000000
1	Pawan	17	90000
2	Mayur	18	80000

```
[33] #iloc - index value based  
df.iloc[0:2]
```

	Name	Age	Package
0	Ankit	16	1000000
1	Pawan	17	90000

```
[36] # 5. Filter dataframe
```



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[36]

5. Filter dataframe

Generate

[37]

df

	Name	Age	Package
0	Ankit	16	1000000
1	Pawan	17	90000
2	Mayur	18	80000
3	Viki	19	70000

[38]

df['Age'] > 17

```
0    False
1    False
2     True
3     True
Name: Age, dtype: bool
```

df[df['Age'] > 17]

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[39] df[df['Age'] > 17]

	Name	Age	Package
2	Mayur	18	80000
3	Viki	19	70000

[40] df_filter = df[df['Age'] > 17]
df_filter

	Name	Age	Package
2	Mayur	18	80000
3	Viki	19	70000

[44] df[(df['Age'] >= 17) & (df['Name'] == 'Mayur')]

	Name	Age	Package
2	Mayur	18	80000

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```
[44] df[(df['Age'] >= 17) & (df['Name'] == 'Mayur')]
```

	Name	Age	Package
2	Mayur	18	80000

```
df.where(df['Age'] >= 17, other= 'NA')
```

```
[45] #6 . Rows & Col : operation(add , update , delete)
```

```
[46] df
```

	Name	Age	Package
0	Ankit	16	1000000
1	Pawan	17	90000
2	Mayur	18	80000
3	Viki	19	70000

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```
#add new col  
df['Team'] = ['CEO', 'HR', 'EMP', 'EMP']  
df
```

[47]

	Name	Age	Package	Team
0	Ankit	16	1000000	CEO
1	Pawan	17	90000	HR
2	Mayur	18	80000	EMP
3	Viki	19	70000	EMP

```
df['Bonus'] = df['Package'] * 0.2  
df
```

[49]

	Name	Age	Package	Team	Bonus
0	Ankit	16	1000000	CEO	200000.0
1	Pawan	17	90000	HR	18000.0
2	Mayur	18	80000	EMP	16000.0
3	Viki	19	70000	EMP	14000.0

add new rows

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add new rows

```
df.loc[len(df)] = ['ABC', 21, 21000, 'IT', 30000]
```

len(df)

[50]

4

update value

```
df.loc[0, 'Package'] = 1000000
```

[51]

	Name	Age	Package	Team	Bonus
0	Ankit	16	1000000	CEO	200000.0
1	Pawan	17	90000	HR	18000.0
2	Mayur	18	80000	EMp	16000.0
3	Viki	19	70000	EMP	14000.0

3

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```
#update using col name  
df.loc[df.Name == 'Ankit', 'Package'] = 9000000  
df
```

[53]

	Name	Age	Package	Team	Bonus
0	Ankit	16	9000000	CEO	200000.0
1	Pawan	17	90000	HR	18000.0
2	Mayur	18	80000	EMp	16000.0
3	Viki	19	70000	EMP	14000.0

```
# Delete
```

```
df.drop(df[df.Name == 'ABC'].index)  
#df.drop(0, axis=1)
```

[54]

	Name	Age	Package	Team	Bonus
0	Ankit	16	9000000	CEO	200000.0
1	Pawan	17	90000	HR	18000.0
2	Mayur	18	80000	EMp	16000.0
3	Viki	19	70000	EMP	14000.0

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[56] `df.drop('Bonus', axis=1)`

	Name	Age	Package	Team
0	Ankit	16	9000000	CEO
1	Pawan	17	90000	HR
2	Mayur	18	80000	EMp
3	Viki	19	70000	EMP

[57] `# sort value
df.sort_values(['Package'])`

	Name	Age	Package	Team	Bonus
3	Viki	19	70000	EMP	14000.0
2	Mayur	18	80000	EMp	16000.0
1	Pawan	17	90000	HR	18000.0
0	Ankit	16	9000000	CEO	200000.0

`df.sort_values(['Package'], ascending=False)`

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0	Ankit	16	9000000	CEO	200000.0
---	-------	----	---------	-----	----------

[59] df.sort_values(['Package'], ascending=False)

	Name	Age	Package	Team	Bonus
0	Ankit	16	9000000	CEO	200000.0
1	Pawan	17	90000	HR	18000.0
2	Mayur	18	80000	EMP	16000.0
3	Viki	19	70000	EMP	14000.0

[1] # 7 . Working with date values

```
df['DOJ'] = ['2024-01-01' , '2024-01-15' , '2024-03-28' , '2024-03-03']
df
```

```
df['DOJ'].dtype
```

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7 . Working with date values

```
df['DOJ'] = ['2024-01-01', '2024-01-15', '2024-03-28', '2024-03-03']
```

```
df
```

```
df['DOJ'].dtype
```

```
df['DOJ'] = pd.to_datetime(df['DOJ'])
```

```
df['DOJ'].dtype
```

```
df1 = df
```

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```
df['DOJ'].dtype
```

```
df1 = df
```

```
df1['DOJ2'] = ['01-01-2025', '02-01-2025']  
df1
```

```
df['DOJ2'].dtype
```

```
df['DOJ2'] = pd.to_datetime(df['DOJ2'], format = '%d-%m-%Y')
```

```
df['DOJ2'].dtype
```

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```
df['DOJ2'] = pd.to_datetime(df['DOJ2'], format = '%d-%m-%Y')
```

```
df['DOJ2'].dtype
```

```
df
```

```
df['DOJ'].dt.year  
df['DOJ'].dt.date  
df['DOJ'].dt.month  
df['DOJ'].dt.day_name() #extractmonth, date, year
```

```
df['Month'] = df['DOJ'].dt.month
```

```
df['DOJ'] + pd.Timedelta(days=90)
```

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```
df['Month'] = df['DOJ'].dt.month
```

```
df['DOJ'] + pd.Timedelta(days=90)
```

```
# Handling Missing Values
```

```
df.isnull()
```

```
import numpy as np
```

```
df.loc[df.Name=='Ankit', 'Package'] = np.null
```

```
df.isnull()
```

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```
import numpy as np
df.loc[df.Name=='Ankit', 'Package'] = np.null
```

```
df.isnull()
```

```
df.isnull().sum
```

```
df.fillna(0)
```

```
# 9. Aggregation and group by
```

```
df['Team'].value_counts()
```

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9. Aggregation and group by

df['Team'].value_counts()

df[df['Month'] == 1].value_counts()

df.groupby('Month')['package'].mean()

df.groupby('Month').agg({'Package' : 'mean' , 'Name' : 'count'})

10. Concatenate and Merge dataframes (Joins)

import pandas as pd

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10. Concatenate and Merge dataframes (Joins)

I

Generate

```
import pandas as pd
df1 = pd.DataFrame({'ID':[1,2,3], 'Name':['A', 'B', 'C']})
df1
```

[5]

	ID	Name
0	1	A
1	2	B
2	3	C

```
df2 = pd.DataFrame({'ID':[1,2,3], 'Score': [88,99,98]})
df2
```

[7]

	ID	Score
0	1	88
1	2	99
2	3	98

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```
df2 = pd.DataFrame({'ID':[1,2,3], 'Score': [88,99,98]})
df2
```

	ID	Score
0	1	88
1	2	99
2	3	98

```
pd.concat([df1, df2], axis=0) #vertical
```

	ID	Name	Score
0	1	A	NaN
1	2	B	NaN
2	3	C	NaN
0	1	NaN	88.0
1	2	NaN	99.0
2	3	NaN	98.0

```
pd.concat([df1, df2], axis=1) #horizontally
```

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1	2	B	NaN
2	3	C	NaN
0	1	NaN	88.0
1	2	NaN	99.0
2	3	NaN	98.0

```
[9] pd.concat([df1, df2], axis=1) #horizontally
```

	ID	Name	ID	Score
0	1	A	1	88
1	2	B	2	99
2	3	C	3	98

```
[10] # Merge =join
```

```
[11] pd.merge(df1, df2, how='inner', on='ID')
```

	ID	Name	Score
--	----	------	-------

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[10] # Merge = join

[11] pd.merge(df1, df2, how='inner', on='ID')

	ID	Name	Score
0	1	A	88
1	2	B	99
2	3	C	98

[12] pd.merge(df1, df2, how='right', on='ID')

	ID	Name	Score
0	1	A	88
1	2	B	99
2	3	C	98

pd.merge(df1, df2, how='inner', left_on='ID', right_on='ID')

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0	1	A	88
1	2	B	99
2	3	C	98

[13]

```
pd.merge(df1, df2, how='inner', left_on='ID', right_on='ID')
```

	ID	Name	Score
0	1	A	88
1	2	B	99
2	3	C	98

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